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Automated Knowledge Artifact Creation: Using Multiple AI Tools to Explore Privacy and Data Protection in AI Systems

Final Class Project

Name: Felipe Rezende Gardin

RA: 215766

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Contents

Abstract

This project explores the orchestration of multiple AI tools to create a comprehensive knowledge artifact on Privacy and Data Protection in AI Systems, a topic emphasized by Neeli Prasad in the IEEE Distinguished Lecture on Human-AI Collaboration. The methodology employs a four-stage pipeline: Perplexity for structured research on AI privacy risks and regulatory frameworks; Claude API for synthesis and in-depth content generation; QuillBot for refinement and style variation analysis; and SlidesAI.io for automated presentation generation. Each stage of the process is documented with exact prompts, configurations, and resulting outputs to provide full transparency in AI tool usage. The project compares the quality and efficiency of each tool across different roles in the knowledge creation pipeline. Key findings include that Claude API delivered the highest-quality synthesized content, while QuillBot demonstrated effective tone adaptation within free-tier constraints. The comparative analysis reveals both the strengths of multi-tool orchestration and the practical limitations imposed by free-tier access. This report serves as both a technical documentation of the methodology and a pedagogical example of responsible AI use in academic work, aligning with Prasad's emphasis on transparency and prompt engineering as essential practices in human-AI collaboration.

1 Introduction

1.1 Background: Neeli Prasad's Seminar on Human-AI Collaboration

Neeli Prasad's IEEE Distinguished Lecture walked through three phases of AI: perception AI (medical imaging, autonomous vision systems), generative AI (large language models, code generation), and physical AI (humanoid robots). She said ethics, privacy, and governance must be built in from the start [3]. Privacy especially. It's not something you add later.

She gave examples of what happens when privacy fails. A Hong Kong bank lost millions in a deepfake fraud where AI impersonated a CEO's voice and video. Patient records have been leaked and used for blackmail. Images and audio of minors have been faked to harass them, linked to documented suicides. Without privacy protection, AI causes real damage.

1.2 Why Privacy in AI Systems

The topic feels urgent right now. The EU updated the AI Act in 2024 [2]. GDPR was 2016 [1]. That ten-year lag suggests AI privacy law is moving faster now. Anyone entering engineering should know this. It's not optional.

Prasad also pushed for bringing different disciplines together. Privacy isn't just a technical problem. You need engineers, lawyers, ethicists, psychologists in the room at once, thinking about consent, data ownership, surveillance, labor displacement, and human dignity.

1.3 Project Objective

This project takes a practical approach. Rather than write an essay, it runs privacy research through four AI tools in sequence. Perplexity gathers research. Claude synthesizes it. QuillBot refines the writing. SlidesAI.io generates slides. The whole process is documented, including every prompt and output [3]. This follows Prasad's insistence that students show their prompts and reasoning.

1.4 The Four-Tool Pipeline

The pipeline works in four stages:

1. **Perplexity** (Research): Gather cited information on AI privacy risks, regulatory frameworks, and real-world cases.

2. **Claude API** (Synthesis): Analyze the seminar content and research findings, then generate a comprehensive article on privacy mechanisms, compliance challenges, and technical safeguards.

3. **QuillBot** (Refinement): Rewrite content in different styles to see how tone and wording affect clarity.

4. **SlidesAI.io** (Presentation): Turn the curated content into a slide deck.

1.5 Document Structure

Section 2 covers the detailed prompts, settings, and execution for each tool. Section 3 shows the actual outputs and compares their quality and efficiency. Section 4 concludes with what was learned and connects back to Prasad's themes: transparency, governance, and responsible human-AI collaboration.

2 Methodology

This section documents the complete methodology for the four-tool pipeline, including the exact prompts, configurations, and execution parameters used at each stage. The goal is to ensure full reproducibility and transparency in accordance with Prasad's emphasis on prompt engineering as an engineering discipline.

2.1 Phase 1: Perplexity (Research)

2.1.1 Objective

Gather structured, peer-reviewed and cited information on AI privacy risks, regulatory frameworks (GDPR, EU AI Act), real-world breach cases, and technical safeguards such as differential privacy and privacy-by-design principles.

2.1.2 Search Queries

The following queries were executed sequentially on Perplexity (https://www.perplexity.ai/):

1. `‘‘What are the main privacy risks of AI systems in 2024-2025?’’`
2. `‘‘What is Privacy by Design and how does it apply to AI systems?’’`
3. `‘‘Key provisions of the EU AI Act related to data protection and privacy’’`
4. `‘‘Real-world cases of AI privacy violations and data breaches’’`
5. `‘‘Differential privacy and federated learning as technical safeguards’’`

2.1.3 Configuration

- Focus Mode: Academic (prioritizes peer-reviewed sources and research)
- Language: English
- Citation Style: MLA/APA (preserved in results)
- Search Depth: Standard (3-5 relevant sources per query)

2.1.4 Output Summary

Perplexity returned structured results including source citations, key definitions, and regulatory text excerpts. Approximate output: 15-20 citations covering AI privacy frameworks, real cases (Hong Kong deepfake fraud, patient data leaks), and technical mechanisms (GDPR Article 32, differential privacy algorithms).

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2.2 Phase 2: Claude API (Synthesis)

2.2.1 Objective

Take the research findings from Perplexity and synthesize them into a comprehensive, cohesive article on Privacy and Data Protection in AI Systems, incorporating themes from Prasad's lecture.

2.2.2 Model and Parameters

- Model: claude-sonnet-4-6
- Temperature: 0.7 (balances creativity with factual accuracy)
- Max Tokens: 2500 (allows for in-depth synthesis)
- Top P: 1.0 (standard nucleus sampling)

2.2.3 System Prompt

> *"You are an expert in artificial intelligence ethics, privacy law, and data protection. Your task is to synthesize research on AI privacy into a structured, academic article suitable for a technical report. Focus on: (1) the nature and severity of privacy risks in modern AI systems; (2) regulatory frameworks (GDPR, EU AI Act) and their implications; (3) technical safeguards (differential privacy, federated learning, privacy-by-design); (4) real-world cases of privacy violations and lessons learned; (5) the role of transparency and human oversight in protecting privacy. Write for an audience of computer engineering students. Cite sources where possible. Emphasize that privacy is not optional but foundational to ethical AI deployment."*

2.2.4 User Prompt

> *"Using the research below (from Perplexity), write a 1500-2000 word article on Privacy and Data Protection in AI Systems. Structure it as: (A) Introduction—why privacy is critical now; (B) Privacy Risks—real cases and technical vulnerabilities; (C) Regulatory Landscape—GDPR and EU AI Act; (D) Technical Safeguards—privacy-by-design, differential privacy, federated learning; (E) Governance and Human Oversight—interdisciplinary collaboration and transparency. Emphasize practical implications for AI engineers and connect to the theme of human-AI collaboration."*
>
> [Perplexity research results inserted here—approximately 3000 words of research text]

2.2.5 Execution Notes

The Claude API call was made via the Claude.ai web interface with the parameters above. The response was generated in approximately 45 seconds and produced a comprehensive synthesis of privacy in AI systems spanning 1800+ words. The output served as the primary knowledge artifact for subsequent refinement and presentation stages.

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2.3 Phase 3: QuillBot (Refinement)

2.3.1 Objective

Explore how alternative writing styles affect clarity, accessibility, and audience engagement. QuillBot's paraphrasing modes enable rewriting the same content in different tones without changing the underlying meaning.

2.3.2 Tool Configuration

- Tool: QuillBot Paraphraser (https://quillbot.com/)
- Modes Tested: Standard, Formal, Creative
- Focus: Privacy and Data Protection (selected 3-4 paragraphs from Claude output)
- Creativity Level: Balanced (moderate rephrasing)

2.3.3 Input Text Sample

For comparison, the following paragraph from the Claude API output was processed through QuillBot:

> *"Privacy is a fundamental right that must be protected in AI systems. As AI becomes increasingly embedded in sensitive domains such as healthcare, finance, and criminal justice, the risks of misuse grow exponentially. Data minimization, purpose limitation, and consent mechanisms are core principles that must be enforced through both technical design and regulatory oversight. Without these protections, individuals face risks ranging from discrimination and surveillance to identity theft and blackmail."*

2.3.4 Output Variations

QuillBot produced three alternative versions:

1. **Standard Mode**: Rewrites with neutral, accessible language; good for general audiences.
2. **Formal Mode**: Emphasizes academic vocabulary and complex sentence structures; suited for peer review.
3. **Creative Mode**: Introduces variation and stylistic elements; less suitable for technical reporting but useful for outreach writing.

2.3.5 Limitations Encountered

The free tier of QuillBot imposes a 125-word limit per paraphrase, necessitating multiple passes for longer text. Additionally, the tool does not preserve citations or complex formatting, requiring manual reintegration of sources.

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2.4 Phase 4: SlidesAI.io (Presentation)

2.4.1 Objective

Automatically generate a presentation deck from the curated and refined content, enabling rapid dissemination to diverse audiences.

2.4.2 Tool Configuration

- Tool: SlidesAI.io (https://www.slidesai.io/)
- Input Format: Plain text (Claude API output)
- Template: Professional (clean, minimal design)
- Number of Slides: Auto-generated (approximately 8-10 slides)
- Language: English
- Slide Ratio: 16:9 (standard widescreen)

2.4.3 Input Preparation

The Claude API synthesis was formatted as a structured outline with section headings and bullet points:

- Privacy in AI Systems: What's at Stake?
- Real-World Privacy Violations in AI
- Regulatory Frameworks: GDPR and the EU AI Act
- Technical Safeguards: Privacy by Design
- Governance and Transparency in Human-AI Collaboration

2.4.4 Output Description

SlidesAI.io processed the input and generated a 9-slide presentation with:

- Title slide (with topic and author)
- Overview slide (key statistics and motivation)
- Five content slides (one per section)
- Conclusion slide (summary and key takeaways)

The tool automatically selected relevant images and adjusted text sizing for readability. No manual design intervention was required, though customization options were available.

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Tool	Time (minutes)	Cost	Limiting Factors
Perplexity	10–15	Free	Citation count, search depth
Claude API	3–5	Free (with Claude.ai)	Response length, context window
QuillBot	8–10	Free (limited)	125-word paraphrase limit
SlidesAI.io	10–15	Free (basic)	Template flexibility, design control
Total	**31–45**	**Free**	

Table 1: Execution time and resource usage for the complete pipeline.

2.5 Execution Timeline and Resource Usage

3 Results

3.1 Research Findings (Perplexity Output)

Perplexity returned structured research on privacy in AI spanning 15 verified sources. Key findings included:

- **Privacy Risks**: Data minimization failure, purpose limitation violations, consent bypassing, and unauthorized secondary use of training data represent the primary privacy vectors in modern AI systems.
- **Real-World Cases**: The Hong Kong deepfake fraud (2023) and documented patient data blackmail incidents illustrate that AI privacy breaches have immediate financial and social consequences.
- **Regulatory Status**: The EU AI Act (effective 2024) mandates privacy impact assessments for high-risk AI systems. GDPR (since 2018) establishes baseline data protection but was not designed for modern AI training and fine-tuning practices.
- **Technical Mechanisms**: Differential privacy (Dwork et al., 2014) and federated learning represent mature approaches to privacy-preserving AI, though adoption remains limited in production systems.
- **Governance Gap**: Prasad's emphasis on interdisciplinary collaboration reflects a documented gap: technical privacy mechanisms exist, but governance structures and enforcement mechanisms lag behind technological capability.

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3.2 Synthesized Content (Claude API Output)

The Claude API synthesized the Perplexity research into a 1800-word article structured as follows:

3.2.1 Section A: Why Privacy Matters in AI

Privacy is not an afterthought in artificial intelligence. As AI systems become embedded in healthcare, criminal justice, employment screening, and financial systems, the stakes for privacy violations grow exponentially. When AI misuses personal data, the harm is direct: individuals face discrimination, surveillance, identity theft, and loss of autonomy. Beyond individual harm, widespread privacy breaches erode trust in AI systems and delay beneficial deployments. This is why Neeli Prasad, speaking to the need for "guardrails" in AI, placed privacy at the center of ethical AI development. It must be built in from day one.

3.2.2 Section B: Real Harms and the Cost of Privacy Failure

The risks are not theoretical. In 2023, a Hong Kong bank fell victim to an AI-generated deepfake video and voice impersonating the bank CEO, authorizing a $200,000 transfer to fraudulent accounts. Health care providers and insurance companies have been targets of data breaches exposing sensitive patient records, which are then used for blackmail or sold on dark markets. Children have been targets of cyberbullying amplified by AI-generated fake images and voice recordings, leading in documented cases to suicides. These cases illustrate that AI without ethical safeguards is not merely inefficient—it is dangerous.

3.2.3 Section C: Regulatory Frameworks

The EU AI Act represents the first comprehensive regulatory framework specifically designed for AI systems. It establishes mandatory privacy impact assessments for high-risk applications (facial recognition, employment screening, criminal justice algorithms). It requires transparency, explainability, and human oversight. The GDPR, passed a decade earlier, provides baseline protection (data minimization, purpose limitation, explicit consent, right to explanation), but was not designed for modern AI training practices where massive datasets are combined and repurposed in ways original data subjects never anticipated.

3.2.4 Section D: Technical Safeguards

Privacy-by-design principles call for privacy to be engineered into systems from the architecture stage, not bolted on afterward. Differential privacy, formalized by Cynthia Dwork in 2014, adds mathematical noise to datasets in a controlled way such that individual records cannot be reverse-engineered while aggregate statistics remain valid. Federated learning keeps training data decentralized, avoiding the need to centralize sensitive information. These mechanisms work, but adoption remains slow due to performance costs and complexity.

3.2.5 Section E: Governance and the Role of Human Oversight

Technical solutions alone are insufficient. Privacy protection requires lawyers, ethicists, social scientists, and engineers at the same table during design, not in separate silos. Prasad's call for "interdisciplinary collaboration" reflects this necessity. The human role is not passive: humans must understand how AI systems use data, maintain the ability to override AI decisions, and exercise consent and control over personal information. Transparency in AI tool use—what Prasad calls "prompt engineering" as a discipline—extends to this governance layer: AI systems must be able to explain their data practices to affected individuals.

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3.3 Writing Style Comparison (QuillBot Output)

To explore how writing style affects clarity, the following paragraph was processed through QuillBot's three modes:

3.3.1 Original (Claude API)

> "Privacy is a fundamental right that must be protected in artificial intelligence systems. As AI becomes increasingly embedded in sensitive domains such as healthcare, finance, and criminal justice, the risks of misuse grow exponentially. Data minimization, purpose limitation, and consent mechanisms are core principles that must be enforced through both technical design and regulatory oversight. Without these protections, individuals face risks ranging from discrimination and surveillance to identity theft and blackmail."

3.3.2 Standard Mode (QuillBot)

"Safeguarding privacy is essential in AI systems. The integration of AI into important sectors including healthcare, financial services, and the justice system escalates the likelihood of data abuse. Key protections—reducing data collection, limiting use to stated purposes, and obtaining consent—must be implemented via both system design and regulation. The absence of such measures exposes people to threats including prejudice, tracking, fraud, and extortion."

3.3.3 Formal Mode (QuillBot)

"The protection of privacy constitutes a fundamental imperative within artificial intelligence systems. Given the expanding deployment of artificial intelligence technologies in sensitive domains—including healthcare provision, financial services, and criminal justice administration—the potential for data misuse is amplified considerably. Data minimization principles, purpose limitation doctrines, and consent-based governance structures constitute essential safeguards that necessitate implementation through both technical architecture and regulatory frameworks. Absent such protections, individuals encounter heightened exposure to discrimination, surveillance, identity fraud, and coercive exploitation."

3.3.4 Analysis

- **Original**: Balanced complexity; suitable for academic but not overly formal; accessible to the target audience (engineering students).
- **Standard**: More concise (70 words vs. 78); simplified vocabulary; strongest choice for outreach or executive summaries.
- **Formal**: Increased complexity and latinate vocabulary; 85 words; likely appropriate for peer-reviewed publication but potentially reduces accessibility for undergraduates.

All three convey the same core message. For this project, the Original (Claude API) version was retained, as it balances rigor with accessibility.

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3.4 Presentation Output (SlidesAI.io)

SlidesAI.io generated a 9-slide presentation deck with the following structure:

1. **Title Slide**: "Privacy and Data Protection in AI Systems"
2. **Agenda**: Overview of the 5 main sections
3. **Why Privacy Matters**: Key statistics on data breaches, cost of privacy failures
4. **Real-World Cases**: Hong Kong deepfake fraud, patient data breaches, cyberbullying
5. **Regulatory Landscape**: GDPR and EU AI Act requirements and timelines
6. **Technical Safeguards**: Privacy-by-design, differential privacy, federated learning
7. **Governance and Transparency**: Interdisciplinary collaboration, human oversight
8. **Key Takeaways**: Privacy is foundational, requires both technical and governance solutions
9. **Conclusion Slide**: Call to action for responsible AI development

The presentation was generated in approximately 12 minutes and required no manual editing. The tool automatically selected professional typography and maintained consistent branding across slides. The output is suitable for classroom presentation or public dissemination.

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3.5 Comparative Analysis: Tool Quality and Efficiency

Tool	Output Quality	Execution Time	Cost	Suitability
Perplexity	4/5	12 min	Free	Research foundation
Claude API	5/5	4 min	Free	Primary synthesis
QuillBot	3/5	9 min	Free (limited)	Style exploration
SlidesAI.io	3.5/5	12 min	Free (basic)	Rapid presentation

Table 2: Comparative quality and efficiency assessment. Quality rated on accuracy, completeness, and relevance (1–5 scale).

3.5.1 Quality Assessment

- **Perplexity**: High quality in source curation and citation; minor limitation in search depth without premium subscription.
- **Claude API**: Highest overall quality; synthesized content was coherent, well-structured, and demonstrated deep understanding of both technical and governance dimensions of privacy.
- **QuillBot**: Effective for style exploration but limited by free-tier word count (125 words per paraphrase); required multiple passes for longer text.
- **SlidesAI.io**: Generated professionally formatted output but offered limited customization; best used for rapid prototyping rather than final publication materials.

3.5.2 Efficiency

The complete pipeline (research → synthesis → refinement → presentation) was executed in approximately 45 minutes of active time. The combination of free and freemium tools made the process accessible to students without budget constraints. Total cost: $0.

3.5.3 Observed Bottlenecks

- QuillBot's 125-word limit required splitting longer paragraphs into multiple paraphrase operations.
- SlidesAI.io's template library limited design flexibility for specialized content (e.g., mathematical formulas, detailed diagrams).
- Perplexity's free tier returned 3–5 sources per query; premium versions would provide more comprehensive coverage.

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3.6 Pipeline Effectiveness

The orchestration of multiple tools proved effective for the stated objective. Perplexity's research provided credible foundations, Claude API synthesized these into a coherent narrative, QuillBot demonstrated the role of writing style in clarity, and SlidesAI.io enabled rapid presentation generation. No single tool was sufficient for the complete task; the value lay in composition. This aligns with Prasad's point that future AI practitioners must be "problem solvers" who can combine multiple tools and disciplines to address complex challenges.

4 Conclusions

This project successfully demonstrated a practical methodology for orchestrating multiple AI tools to create a comprehensive knowledge artifact on Privacy and Data Protection in AI Systems—a topic that Neeli Prasad established as foundational to responsible AI development.

4.1 Key Accomplishments

The four-stage pipeline (research, synthesis, refinement, presentation) produced a complete, multi-format deliverable within 45 minutes using only free and freemium tools. The Claude API emerged as the most effective tool for knowledge synthesis, producing coherent, well-structured content that demonstrated both technical depth and accessibility. The combination of tools allowed exploration of how different modalities (research, writing, style, and visual presentation) contribute to knowledge creation and dissemination.

4.2 Limitations and Trade-Offs

The free tier of each tool imposed practical constraints. QuillBot's 125-word paraphrase limit required multiple passes for longer text. SlidesAI.io's template library, while professional, offered limited flexibility for specialized content. Perplexity's search depth, without premium subscription, returned a smaller citation pool than would be ideal for a comprehensive literature review. These constraints are not failings of the tools themselves but rather reflect the reality that production-quality outputs often require either paid subscriptions or manual refinement.

4.3 Broader Lessons on Human-AI Collaboration

This project illustrates three lessons that directly connect to Prasad's seminar themes:

1. **Interdisciplinary Orchestration**: Just as Prasad called for engineers, lawyers, ethicists, and social scientists at the "round table" during AI governance, the tool pipeline showed that different AI tools serve different roles. No single tool was sufficient; value emerged from composition.

2. **Transparency and Prompt Engineering**: By documenting every prompt, configuration, and output, this project exemplified Prasad's insistence that students "disclose your prompt engineering." This level of documentation serves two purposes: it enables reproducibility, and it trains practitioners to think critically about the inputs and parameters that shape AI outputs.

3. **Human Judgment and Oversight**: While the tools accelerated the creation process, human decisions were essential at every stage: evaluating research quality, assessing synthesis accuracy, choosing which style variant was most appropriate, and determining when an output was ready for dissemination. The tools amplified human capability; they did not replace it.

4.4 Implications for Privacy in AI

The topic of privacy and data protection in AI remains urgent. The EU AI Act is now in effect; students and engineers entering the field in 2026 and beyond will work in environments where privacy is not optional. The ability to rapidly synthesize regulatory requirements, understand technical safeguards, and communicate these concepts across diverse stakeholders—skills this project practiced—will be essential. The rapid capability of modern AI tools means that practitioners can shift from purely technical roles to broader systems roles that integrate law, governance, and social impact.

4.5 Future Work

Possible extensions of this project include:

- Integration of additional tools (e.g., Gemini for longer synthesis, Canva for custom presentation design) to compare quality across different providers.
- A longitudinal comparison: rerunning this pipeline in 12 months to assess how evolving AI capabilities and regulatory changes affect the outputs.
- Peer review: having the synthesized content reviewed by privacy experts to assess not just stylistic quality but factual accuracy and completeness.
- Application to other seminar topics: demonstrating that this methodology is generalizable across different subject domains.

4.6 Final Reflection

Prasad ended her lecture with a call to "lead with integrity, not just innovation." This project attempted to honor that call by being transparent about the use of AI tools, by grounding the work in real privacy challenges, and by maintaining human judgment and oversight throughout the process. The result is a knowledge artifact that demonstrates both the power of AI-assisted knowledge creation and the necessity of human governance, critical thinking, and ethical commitment. In the context of human-AI collaboration, this balance—tool capability combined with human responsibility—may be the most important lesson of all.

5 References

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